

Bluestock

Mutual Fund Analytics Platform

Final Project Report

End-to-end analytics: ETL · SQLite warehouse · performance & risk analytics · forward-looking simulations · Tableau dashboard

40 schemes · 64,320 daily NAVs · 32,778 transactions

Prepared by

Dheeraj Reddy Thumma

Data Analyst — Internship Capstone · Status: COMPLETE

1. Executive Summary

- The Bluestock Mutual Fund Analytics Platform is a complete, reproducible data-engineering and analytics project covering 40 Indian mutual fund schemes across 10 fund houses (Equity and Debt; 12 sub-categories).
- A Python ETL pipeline cleans and validates the raw datasets and loads them into a star-schema SQLite warehouse (2 dimensions + 5 fact tables). Four analytics notebooks deliver EDA, performance, risk, and forward-looking simulation analytics; a recommendation engine serves risk-profiled picks; and a 4-page Tableau dashboard presents results to stakeholders.
- Headline result: Mirae Asset Large Cap Fund leads on every risk-adjusted measure (Sharpe 1.07, Sortino 1.49, composite score 85.9/100). Small-cap funds carry the highest tail risk and deepest drawdowns (SBI Small Cap: -52.6%). A Monte Carlo projection and a Markowitz efficient frontier extend the platform from descriptive to forward-looking. Data quality is high — 40/40 AMFI codes validated, zero invalid/duplicate rows after cleaning.

Key numbers

Schemes	40	Fund houses	10
Daily NAV records ..	64,320	Transactions	32,778
Benchmark records ..	8,050	Date range	2022-01-03 -> 2026-05-29
Analytics notebooks	4	Analytics charts ...	34
Top fund (score) ...	Mirae Asset Large Cap Fund (85.9/100)		

2. Data

Ten source datasets were ingested from data/raw/ and profiled (rows x columns, missing values):

dataset	rows	cols	missing
01_fund_master.csv	40	15	0
02_nav_history.csv	46,000	3	0
03_aum_by_fund_house.csv	90	5	0
04_monthly_sip_inflows.csv	48	6	12
05_category_inflows.csv	144	3	0
06_industry_folio_count.csv	21	6	0
07_scheme_performance.csv	40	19	0
08_investor_transactions.csv	32,778	13	0
09_portfolio_holdings.csv	322	8	0
10_benchmark_indices.csv	8,050	3	0

- Universe: 40 schemes, 10 fund houses, 2 categories (Equity, Debt), 12 sub-categories. NAV history spans 2022-01-03 to 2026-05-29.

3. ETL Pipeline

A fail-fast Python pipeline (run_pipeline.py) executes three stages: Validation, Cleaning, and Database Build.

Cleaning results

- NAV history: 46,000 raw rows -> 64,320 cleaned. 0 duplicates, 0 invalid NAVs; 18,320 weekend/holiday dates added by per-scheme forward-fill.
- Investor transactions: 32,778 rows; 0 duplicates, 0 invalid amounts/types/KYC, 0 missing values after standardisation.

Warehouse (SQLite star schema)

dim_fund	40	fact_performance	40
dim_date	1,608	fact_aum	90
fact_nav	64,320	fact_benchmark	8,050
fact_transactions	32,778		

4. EDA & Advanced Analytics Findings

Distribution & behaviour

- Daily returns: mean +0.045% (~+11.9% annualised), median 0.0%; fat-tailed (excess kurtosis +3.10). ~39% up / 33% down / 29% flat days.
- Universe is 85% equity (34/40). Investors: 66% male, 66% T30 metros. SIPs are 60% of 32,778 transactions.

Risk & concentration

- Tail risk spans ~93x: ABSL Small Cap CVaR-95 -3.0% vs ICICI Liquid -0.03%.
- SIP continuity: only 30 of 1,362 investors 'Healthy' (1,332 At Risk).
- Diversification (HHI): UTI Mid Cap most diversified (8.1 effective sectors); Axis Bluechip most concentrated (3.4).

5. Performance Analysis

Risk-adjusted leaders

	Sharpe	value
Mirae Asset Large Cap Fund		1.068
Kotak Flexicap Fund		0.966
Mirae Asset Tax Saver Fund		0.919

Growth (3-year CAGR)

Axis Midcap 35.1% · Mirae Large Cap 34.0% · ICICI Bluechip 32.5%

Composite fund scorecard (0-100)

1. Mirae Asset Large Cap	85.9	4. HDFC Mid-Cap Opp. ...	80.6
2. ICICI Pru Midcap	82.4	5. ICICI Pru Bluechip ..	78.9
3. Kotak Flexicap	81.5		

Risk extremes

- Deepest drawdown: SBI Small Cap -52.6% (1,228 days underwater).

6. Bonus Analytics — Forward-Looking Simulations

6.1 Monte Carlo NAV Projection

- Mirae Asset Large Cap Fund (top scorecard fund) projected 5 years forward via 1,000 simulated paths (daily returns $\sim$ Normal(μ , σ)).
- From NAV Rs 230.27: median Rs 585 in 5 years (median CAGR +20.5%), with a wide P5-P95 band of Rs 391 to Rs 909 — outcomes expressed as a range, not a single point forecast.

6.2 Markowitz Efficient Frontier

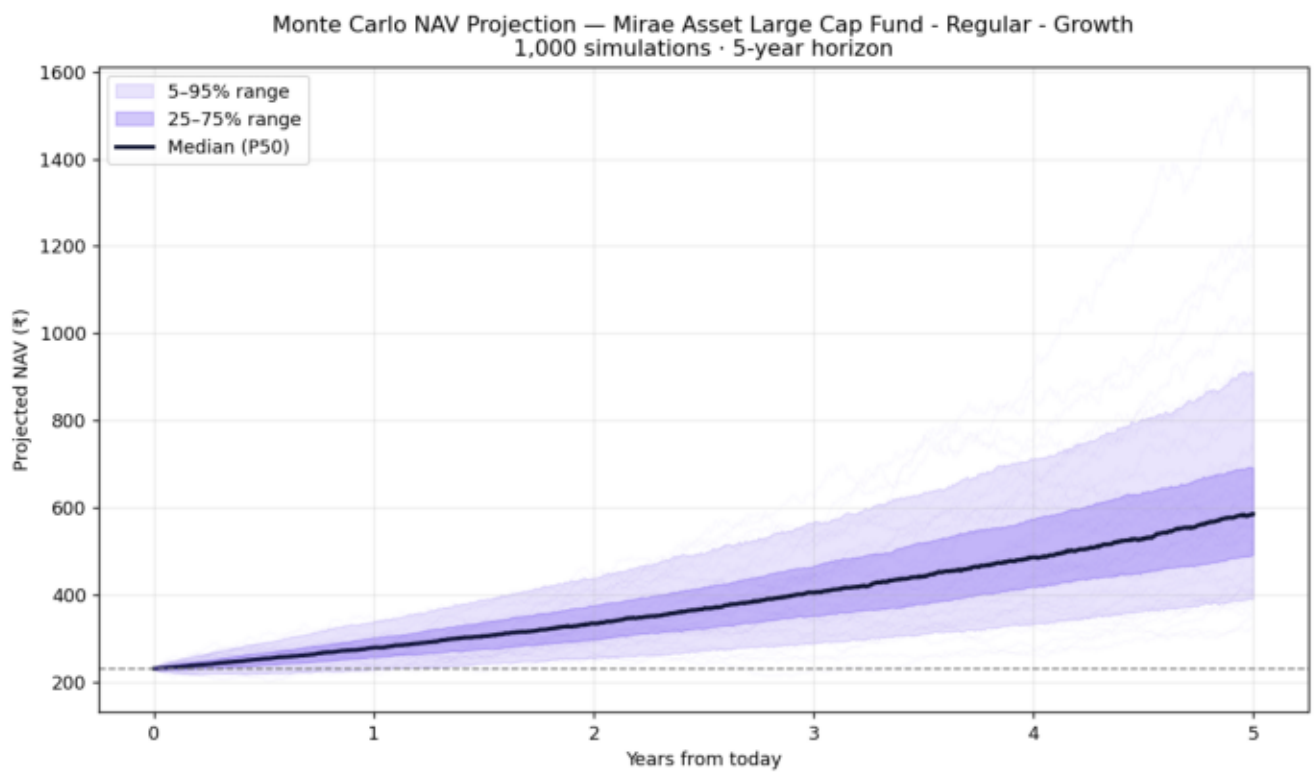
- Long-only optimisation across the top-5 scorecard funds (20,000 simulated portfolios + SLSQP frontier).

Max-Sharpe portfolio: return 18.7%, volatility 5.9%, Sharpe 3.18

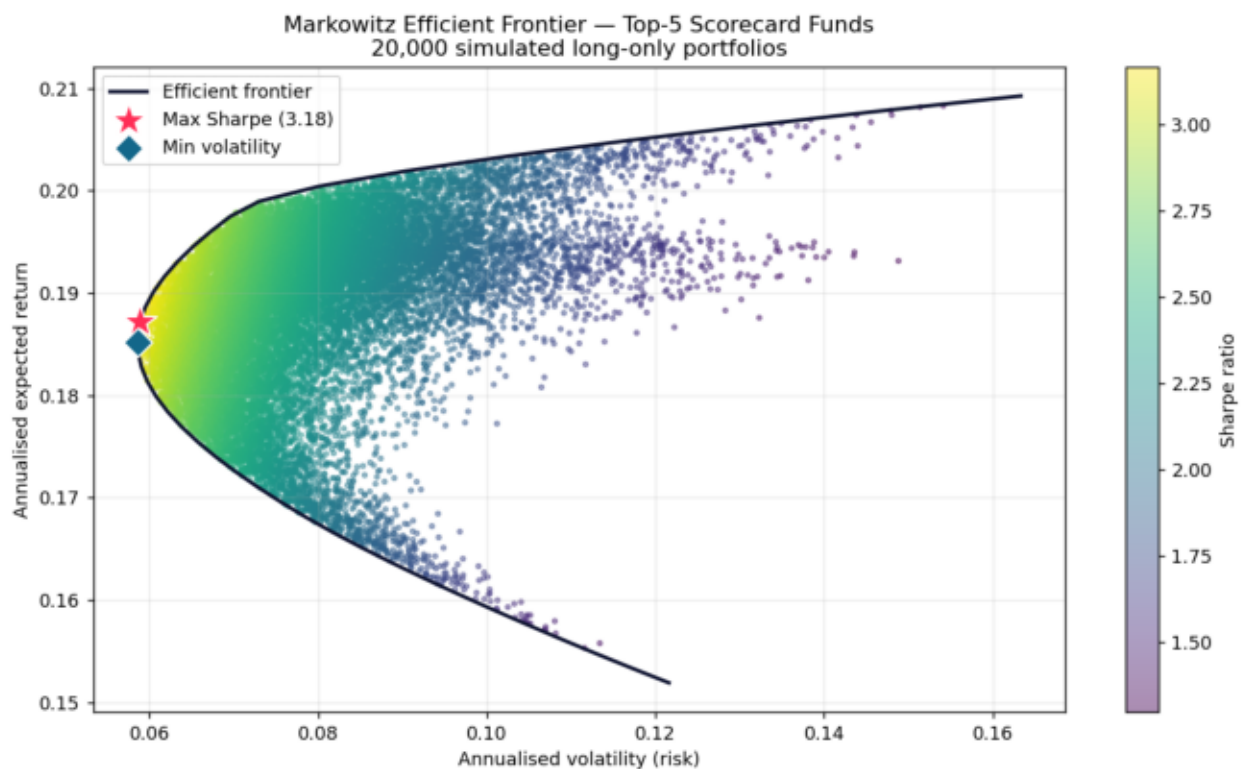
Weights: Mirae 26% ICICI Bluechip 22% Kotak 21% ICICI Midcap 16% HDFC 15%

- Caveat: the five funds show near-zero correlation (a synthetic-dataset trait), so diversification looks unusually strong and Sharpe is inflated. The methodology is sound; the magnitudes are illustrative for this data.

6. Bonus Analytics — Simulation Charts



6.1 Monte Carlo — 5-year NAV projection with P5-P95 bands



6.2 Markowitz Efficient Frontier — top-5 funds

7. Dashboard — Page 1: Industry Overview

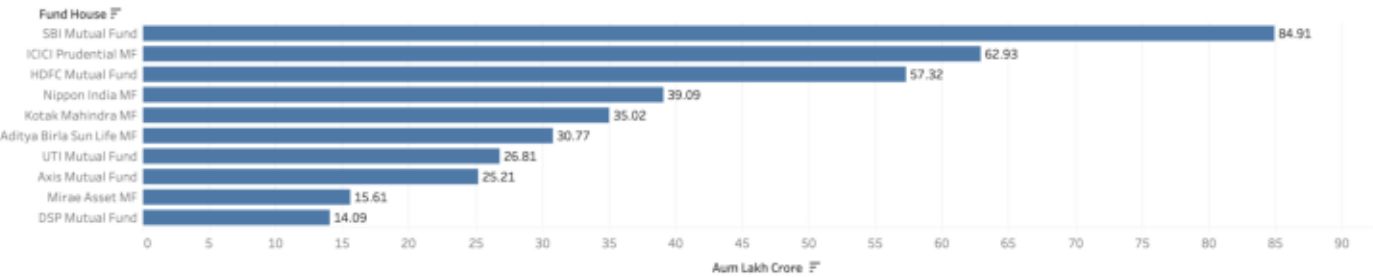
Bluestock Mutual Fund Analytics — Industry Overview

KPI SIP Inflow	KPI Folios	KPI Schemes	KPI Total AUM
31,002	26.12	40	62.74

Industry AUM Trend



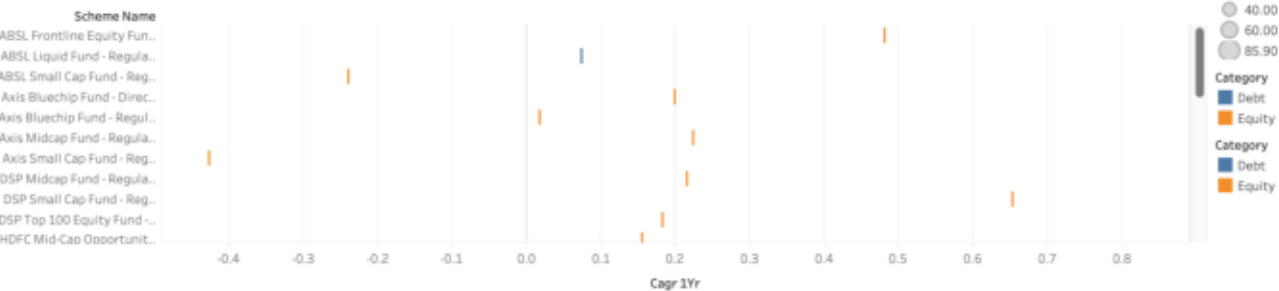
Top Fund Houses by AUM



7. Dashboard — Page 2: Fund Performance

Bluestock Mutual Fund Analytics — Fund Performance

Fund CAGR Comparison



Fund Scorecard

Category	Scheme Name	Fund Sc..	Overall ..	Return ..	Sharpe ..
Debt	ABSL Liquid Fund - Regula..	31.15	33.00	31.00	40.00
	HDFC Short Term Debt Fu..	31.41	32.00	34.00	37.00
	ICICI Pru Liquid Fund - Reg..	37.05	28.00	29.00	38.00
	Kotak Liquid Fund - Regula..	36.41	29.00	30.00	39.00
	Nippon India Gilt Securiti..	34.23	30.00	33.00	36.00
	SBI Magnum Gilt Fund - R..	33.33	31.00	32.00	35.00
Equity	ABSL Frontline Equity Fun..	66.92	10.00	9.00	10.00
	ABSL Small Cap Fund - Re..	20.51	37.00	39.00	27.00
	Axis Bluechip Fund - Direc..	46.92	22.00	16.00	28.00
	Axis Bluechip Fund - Regula..	21.41	36.00	36.00	30.00
	Axis Midcap Fund - Regula..	77.69	6.00	1.00	9.00
	Axis Small Cap Fund - Reg..	15.38	38.00	40.00	32.00
	DSP Midcap Fund - Regula..	65.90	13.00	11.00	6.00
	DSP Small Cap Fund - Reg..	66.28	11.00	10.00	12.00
	DSP Top 100 Equity Fund ..	42.05	24.00	21.00	25.00

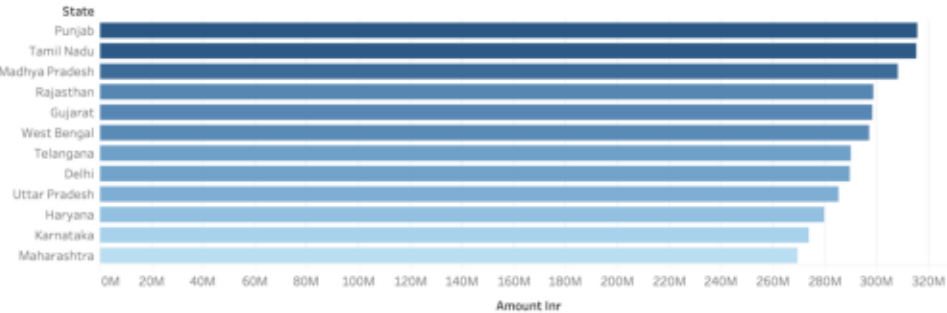
Return vs Risk



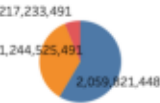
7. Dashboard — Page 3: Investor Analytics

Investor Analytics

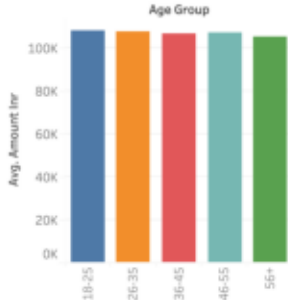
State Transaction Amount



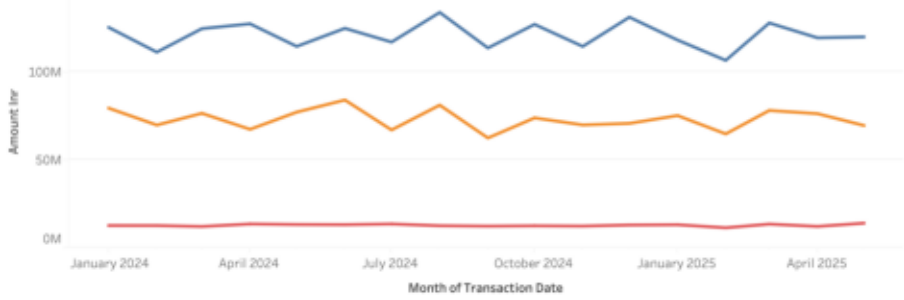
Transaction Type Split



Age Group vs SIP Amount



Monthly Transaction Trend



7. Dashboard — Page 4: SIP & Market Trends

Bluestock Mutual Fund Analytics — SIP & Market Trends

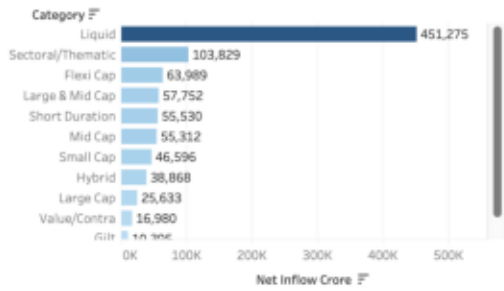
SIP Inflow Trend



Category Inflow Heatmap

Category	Month											
	2024-04	2024-05	2024-06	2024-07	2024-08	2024-09	2024-10	2024-11	2024-12	2025-01	2025-02	2025-03
ELSS	466	553	472	471	499	537	537	571	521	516	437	500
Flexi Cap	4,947	5,529	4,478	4,869	5,562	5,397	6,004	6,111	4,654	5,603	6,068	4,767
Gilt	784	836	864	959	952	925	898	704	831	744	942	956
Hybrid	2,955	3,487	3,163	3,291	3,684	3,015	3,314	3,264	3,538	2,967	3,360	2,830
Large & Mid Cap	4,214	4,368	4,610	5,023	5,411	4,528	4,581	5,556	4,878	4,816	5,524	4,243
Large Cap	2,413	2,076	2,519	2,574	1,940	1,879	2,255	1,870	1,923	2,025	1,925	2,234
Liquid	37,537	41,872	40,486	34,643	41,952	35,308	39,091	40,506	34,933	33,892	32,374	38,681
Mid Cap	3,897	5,300	5,047	4,548	3,899	4,960	4,106	4,336	5,023	4,316	4,819	5,061
Sectoral/Themat...	8,052	8,354	10,030	9,896	8,360	8,518	7,680	7,397	9,820	7,893	9,215	8,614
Short Duration	4,400	4,833	4,321	4,170	4,658	5,327	4,675	5,316	4,159	4,752	4,033	4,886
Small Cap	3,533	4,092	3,535	3,582	3,376	4,137	4,444	3,256	4,249	4,554	3,534	4,304
Value/Contra	1,328	1,361	1,386	1,582	1,308	1,334	1,595	1,436	1,414	1,339	1,400	1,497

Top Categories by Inflow



8. Recommendations

The recommendation engine (src/recommender.py) serves the Top-3 Sharpe-ranked funds per risk profile.

Profile	Rank	Fund	Sub-category
LOW RISK	1	SBI Magnum Gilt Fund	Gilt
LOW RISK	2	Nippon India Gilt Securities Fund	Gilt
LOW RISK	3	HDFC Short Term Debt Fund	Short Duration
MODERATE RISK	1	Mirae Asset Large Cap Fund	Large Cap
MODERATE RISK	2	Kotak Flexicap Fund	Flexi Cap
MODERATE RISK	3	SBI Bluechip Fund	Large Cap
HIGH RISK	1	ICICI Pru Midcap Fund	Mid Cap
HIGH RISK	2	DSP Midcap Fund	Mid Cap
HIGH RISK	3	HDFC Mid-Cap Opportunities Fund	Mid Cap

- Moderate-risk investors map to the strongest risk-adjusted equity funds; low-risk to debt/gilt funds; high-risk to mid-cap funds (small-caps excluded given severe drawdowns).
- The efficient frontier (Section 6.2) complements this by suggesting an optimal multi-fund blend rather than single picks.

9. Limitations

- 5-year CAGR is unavailable: NAV history starts in 2022 (~4.4 years), so 5Y figures are NaN and only 1Y/3Y growth is reliable.
- Point-in-time performance: fact_performance is a single snapshot (no date), so alpha/beta/ratios reflect one reporting period.
- Synthetic-data correlation: the funds show near-zero cross-correlation, which inflates the efficient-frontier Sharpe and understates portfolio volatility (Section 6.2). Methodology transfers to real data; magnitudes are illustrative.
- SIP-continuity skew: the 1,332 'At Risk' figure reflects a strict gap threshold over a short, partial window rather than confirmed mass churn.
- Cohort coverage: only 2024 and 2025 cohorts exist (2025 partial) — trends are directional only.
- Recommendation scope: picks rank purely on Sharpe within a sub-category; expense ratio, fund size and manager tenure are not yet factored in.